MEMS 2023 – SOLUTIONS

Question 1

a) The electrostatic force generated by a parallel plate actuator is $F_e = -\varepsilon_0 A V^2/(2g^2)$, where A is the plate area and g is the gap between the plates. (This result may be derived or simply quoted.) At constant maximum electric field, V/g remains constant, so F_e scales as $A \propto L^2$.

At cm scale and larger, the maximum field is limited by avalanche breakdown, and it is reasonable to assume that the threshold field for this is constant. At smaller scales, the density of gas molecules is insufficient to support avalanche breakdown, and the maximum field increases with decreasing size. At the scale of a few microns, the maximum field becomes constant again, but at a much higher level (set by field emission). [6]

b)



[2]

The bending moment at a point that is x from the LH end is the moment due to loads and couples acting on the beam section to the right of that point, so:

$$M = P(L - x) - C = P(\frac{L}{2} - x)$$

where we have used the fact that $C = PL/2$ which is required for static equilibrium. [2]

The strain at a distance y above the neutral axis is $\varepsilon = -y/R$ where R is the local radius of curvature. But $M = EI/R$ so we can write the strain as:

$$\varepsilon = -\frac{My}{EI} = -\frac{Py}{EI}(\frac{L}{2} - x)$$ [3]

The maximum strain occurs at two positions: $y = \frac{h}{2}, x = L$ and $y = -\frac{h}{2}, x = 0$, and the strain at these points is:

$$\varepsilon_{max} = \frac{PhL}{4EI} = \frac{3v_L h}{L^2}$$

where we have used the fact that the end deflection is $v_L = \frac{PL^3}{12EI}$ (can be quoted). [3]

c) The generalised Hooke's law equations are:

$$E\varepsilon_x = \sigma_x - \upsilon\sigma_y - \upsilon\sigma_z$$

$$E\varepsilon_y = \sigma_y - \upsilon\sigma_z - \upsilon\sigma_x$$

$$E\varepsilon_z = \sigma_z - \upsilon\sigma_x - \upsilon\sigma_y$$

The biaxial modulus is the ratio of stress to strain when there is isotropic plane stress, for example if $\sigma_x = \sigma_y \neq 0$ and $\sigma_z = 0$. Taking this particular example, we see that the first two equations become identical and of the form:

$$E\varepsilon = \sigma(1 - \upsilon)$$

The ratio of stress to strain, which is the biaxial modulus, is then $\tilde{E} = \frac{E}{1-\upsilon}$.

The biaxial modulus is used in situations involving thermally induced stress/strain which has the required symmetry because of the isotropic nature of thermal exansion. [6]

d) Considering, for example, the (111) and (100) planes, the unit normals are $\hat{\boldsymbol{n}}_{\mathbf{111}} = \frac{1}{\sqrt{3}}(1,1,1)$ and $\hat{\boldsymbol{n}}_{\mathbf{100}} = (1,0,0)$. The angle between them is therefore:

$$\theta = \cos^{-1}(\hat{\boldsymbol{n}}_{\mathbf{111}} \cdot \hat{\boldsymbol{n}}_{\mathbf{100}}) = \cos^{-1}\left(1/\sqrt{3}\right) = 54.7°$$

It is clear that the same result will apply to all pairs of {111} and {100} planes.

Considering the case of a [110]-oriented wafer, the following {111} planes will all be normal to the surface: $(1, \bar{1}, 1), (1, \bar{1}, \bar{1})$. This follows because the normal vectors for these planes will have zero dot product with (1,1,0). However, the planes $(1,1,1)$ and $(1,1, \bar{1})$ will NOT be normal to the surface.

The angle between $(1, \bar{1}, 1)$ and $(1, \bar{1}, \bar{1})$ is $\cos^{-1}(1/3) = 70.5°$, so the angles at the corners of a limiting etched cavity will be 70.5° and 109.5°. [6]

e) The main actuation mechanisms are: electrostatic (ES), electrothermal (ET), piezoelectric (PE) and electromagnetic (EM).

ES and ET actuators are easiest to implement because they can be fabricated in a single mechanical layer. EM actuators are most difficult because they require magnetic materials and tend to be geometrically more complex if they include an efficient magnetic circuit.

ES, PE and EM actuators are all fast. ET actuators are slower because they are limited by the thermal time constant of the structure.

ES and EM are limited to relatively low force levels by breakdown and Joule heating respectively. PE and ET actuators can generally develop larger forces. [6]

f) The Reynolds number gives information about the flow regime and, in particular, whether the flow is laminar or turbulent. It measures the ratio between inertial and viscous forces. For flow in a pipe, laminar flow is expected for $Re < \sim 2300$, while for $Re > \sim 3000$ the flow is expected to show turbulence.

The Reynolds number formula is: $Re = \rho UD/\mu$

where ρ is the fluid density, U is the flow speed, D is a characteristic length (e.g. pipe diameter) and μ is the fluid viscosity.

Typical numbers for U and D in a microfluidic device might be 5 mm/s and 100 μm. With ρ = 1000 kg/m^3 and μ = 1 mPa.s we get Re = 0.5 which is very low and well into the laminar regime.

When the flow is laminar mixing cannot occur by churning of the fluid and has to rely on diffusion. Increasing the contact area between the fluids to be mixed and reducing the distance over which diffusion needs to occur speeds up the mixing process. This leads, for example, to designs that divide each flow into an array of narrower sub-flows and then merge the sub-flow arrays so that they are interleaved. [6]

Question 2

a) The suspension stiffness is twice that of a single flexure i.e. $k = 24EI/L^3 = 2Ew^3h/L^3$. With E = 169 GPa, w = 5 μm, h = 10 μm and L = 250 μm, this gives k = 27 N/m.

The resonant frequency is $\omega_0 = \sqrt{k/m} = \sqrt{\frac{k}{\rho Ah}}$ where ρ is the density and A is the proof ass area. With k = 27 N/m, ρ = 2330 kg/m^3, A = 0.12 mm^2 and h = 10 μm this gives ω_0 = 98,268 rad/s or 15.64 kHz. [8]

b) According to the parallel plate model, the capacitance between adjacent electrode fingers is $\varepsilon_0 lh/g$, so the total capacitance between terminal A and ground is $C_A \approx N\varepsilon_0 lh/g$. With N = 50, l = 100 μm, h = 10 μm, g = 1 μm this gives $C_A \approx$ 0.44 pF.

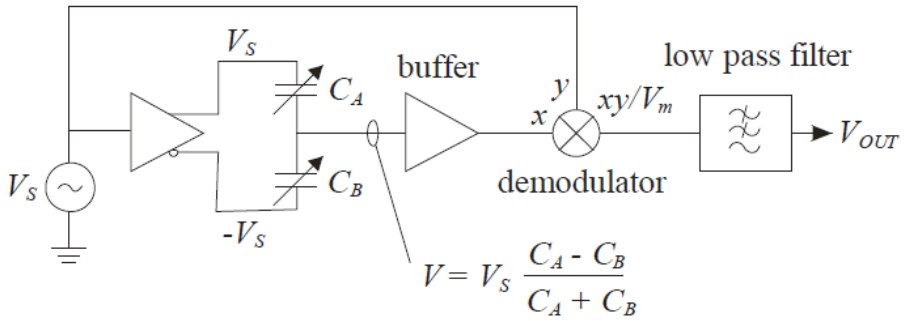
Fringing will make actual capacitance higher, as will parasitic capacitance through substrate, so estimate is expected to be on low side. [8]

c) The differential capacitance under applied acceleration is:

$$C_A - C_B = N\varepsilon_0 lh\left\{\frac{1}{g+x} - \frac{1}{g-x}\right\} \approx -\frac{2N\varepsilon_0 lh}{g^2}x = -2C_A\frac{x}{g}$$

where x is the displacement of mass relative to substrate (+ve displacement to right). For steady state acceleration a, $x = -a/\omega_0^2$, so the steady state displacement at an acceleration of +10 ms^{-2} is x = –1.04 nm. The corresponding differential capacitance is $C_A - C_B \approx$ 0.92 fF. [8]

d) A possible readout scheme would be:



In this scheme, C_A and C_B form a potential divider, the ends of which are driven by complementary clock/carrier signals. The AC signal at the junction between the capacitors will then be proportional to the differential capacitance. This signal is amplified and passed through a coherent demodulator to generate a baseband output voltage. [6]

Question 3

a) We need to show that the given function satisfies $\nabla^2(\nabla^2 w) = \frac{p}{D}$, with p a constant, and also the relevant boundary conditions. Now:

$$\nabla^2(w) = \frac{1}{r}\frac{\partial}{\partial r} r \frac{\partial}{\partial r} w_0 \left(1 - \frac{r^2}{a^2}\right)^2 = \frac{1}{r}\frac{\partial}{\partial r}\left\{\frac{-4w_0}{a^2} r^2 \left(1 - \frac{r^2}{a^2}\right)\right\} = \frac{-8w_0}{a^2}\left(1 - \frac{2r^2}{a^2}\right)$$

So:

$$\nabla^2(\nabla^2 w) = \frac{1}{r}\frac{\partial}{\partial r} r \frac{\partial}{\partial r}\left\{\frac{-8w_0}{a^2}\left(1 - \frac{2r^2}{a^2}\right)\right\} = \frac{1}{r}\frac{\partial}{\partial r}\left\{\frac{32w_0 r^2}{a^4}\right\} = \frac{64w_0}{a^4}$$

The given function is therefore a solution of $\nabla^2(\nabla^2 w) = const$ as required. [6]

The boundary conditions for a built-in circular plate, i.e. $w = 0, \frac{\partial w}{\partial r} = 0$ for $r = a$, are also satisfied. [2]

Substituting for $\nabla^2(\nabla^2 w)$ from above in the bending equation we have $\frac{64w_0}{a^4} = \frac{p}{D}$, so:

$$w_0 = \frac{a^4 p}{64D}$$

[2]

b)

i) A possible fabrication sequence would be:

- Top side lithography to define disc and perforations in mech layer 2
- RIE or DRIE to transfer pattern into mech layer 2
- Liquid or vapour phase sacrificial oxide etch
- Bottom side lithography to define recess
- DRIE through handle wafer
- RIE to remove oxide from underside of membrane [6]

ii) With the given assumptions the capacitance will be $C = \frac{\varepsilon_0 \pi a^2}{g} = 3.48$ pF. [3]

iii) Making use of the hint, the shift in capacitance for an element of area *dA* at radius *r* will be:

$$-\frac{\varepsilon_0}{g^2} w_0 \left(1 - \frac{r^2}{a^2}\right)^2 dA$$

The total shift in capacitance will therefore be:

$$\Delta C = -\frac{\varepsilon_0}{g^2} w_0 \int_0^a \left(1 - \frac{r^2}{a^2}\right)^2 2\pi r dr = -\frac{\varepsilon_0}{g^2} w_0 \frac{\pi a^2}{3}$$

Substituting for w_0 from part a) and using the relation $D = \frac{Eh^3}{12(1-\upsilon^2)}$, the required form is obtained.

[8]

iv) With a = 0.5 mm, h = 10 µm, g = 2 µm, E = 160 GPa, ν = 0.22 and p = 1 kPa, the equation derived in part b)iii) gives:

$$\Delta C \approx -\frac{\pi}{16} \frac{\varepsilon_0 a^6}{g^2 h^3} \frac{1-\nu^2}{E} p = 0.4 \text{ fF}$$

[3]

Question 4

a) The transverse stiffness of a flexure is $k_T = 12EI/L^3$ while the axial stiffness is $k_A = EA/L$. Here L is the beam length and A is its cross-sectional area. (These results may be derived or simply quoted.) The second moment of area for in-plane deflection is given by $I = w^3h/12$, so the transverse stiffness can also be expressed as $k_T = Ew^3h/L^3$. The ratio of the stiffnesses is therefore:

$$\frac{k_T}{k_A} = \frac{Ew^3h}{L^3}\frac{L}{Ewh} = \frac{w^2}{L^2}$$

as required. [8]

b) The problem with using unconstrained expansion is that the displacements obtained are very small for any reasonable temperature rise. On the other hand, an actuator based on unconstrained expansion is very stiff. By using frustrated expansion, stiffness can be traded for displacement, allowing larger displacements to be achieved. [4]

c) For the system to be in mechanical equilibrium, the reactions at the anchors must lie along the *x*-direction. The resultant of P_A and P_T must therefore have zero *y*-component and this leads to the first relation.

From symmetry we know that the point where the two flexures meet must move only in the *y*-direction. The resultant of the displacement vectors u_A and u_T must therefore have zero *x*-component and this leads to the second relation. [4]

Taking into account both the loads and the thermal strain, the displacements will be:

$$u_A = L\varepsilon_{th} - P_A/k_A \quad ; \quad u_T = P_T/k_T$$

Also we know that: $\Delta y = u_T \cos\theta + u_A \sin\theta.$

Combining these relations with the given relations $P_T = P_A \tan\theta$ and $u_A = u_T \tan\theta$, and after some manipulation, the required result is obtained. [8]

d) Differentiating the expression derived in part c) it is found that the function has a single maximum which occurs at:

$$u = \sqrt{\frac{\alpha}{1-2\alpha}} \approx \sqrt{\alpha}$$

where the approximation can be applied because $\alpha \ll 1$. So, we have $\tan\theta \approx w/l$, and since the angle is small it follows that $\theta \approx w/l$. [6]